MIL-C-2069E (GL)
31 December 1968
SUPERSEDING
MIL-C-2069D (GL)
9 November 1964

MILITARY SPECIFICATION

CLOTH, FLEECE, COTTON WARP AND WOOL-NYLON

FILLING: LINING, 15-OUNCE, SHRINK RESISTANT

1. SCOPE

1.1 This specification covers one type of cotton warp and wool-nylon filling cloth.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

CCC-T-191 - Textile Test Methods
PPP-P-1132 - Packaging and Packing of Woolen, Worsted and Wool Blend (Synthetic Fiber; Cotton) Fabrics

STANDARDS

FEDERAL

Fed. Std. No. 4 - Glossary of Fabric Imperfections

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. DEPARTMENT OF AGRICULTURE

Methods of Test for Grades of Wool

(Application for copies should be addressed to U. S. Department of Agriculture, Agricultural Marketing Service, Washington, D. C. 20402)

Rules and Regulations Under the Wool Products Labeling Act of 1939

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402)

3. REQUIREMENTS

3.1 Standard sample.- The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced. (see 6.3).

* 3.2 First article.- Unless otherwise specified (see 6.2), before production is commenced a 1/4 yard sample full width of the untreated cloth and 3-1/2-yard sample full width of the finished cloth shall be submitted or made ready for the contracting officer or his authorized representative for inspection as specified in 4.2. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured by the supplier in the same facilities to be used for the manufacture of the production item.

3.3 Material.-

3.3.1 Fiber.-

3.3.1.1 Wool.- The wool blend used in the filling yarn shall be fleece or pulled sheep's wool (or both) made from the following grades of wool:

(a) Not less than 40 percent, not lower than 56's U. S. Standard.
(b) Not more than 40 percent, not lower than 50's U. S. Standard.
(c) Not more than 20 percent of reprocessed wool, not lower than 48's U. S. Standard, obtained from worsted spun material.

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With the exception of the reprocessed wool specified, the use of any wool manufacturing by-products other than those produced by the supplier's own current production of this cloth is prohibited.

3.3.1.2 Nylon.- The fiber shall be staple nylon of suitable length and denier to conform to the requirements of this specification (see 4.3.1.2).

3.3.2 Yarn.-

3.3.2.1 Warp yarn.- The warp yarn shall be made of carded cotton and shall be spun and twisted into a 2-ply yarn (see 4.3.1.2).

3.3.2.2 Filling yarn.- The filling yarn shall be a blend of not less than 80 percent of the specified wool blend and not less than 10 percent nor more than 20 percent staple nylon based on the dry weight when tested as specified in 4.3.1.2. The stock shall be blended, carded and spun into a single yarn on the woolen system.

* 3.4 Color.- The color may be obtained by stock or piece dyeing. The color of the finished cloth shall match the standard sample in Olive Green 104 (see 6.4). The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid on the cotton warp yarn is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4. When a standard sample is not available, the dyed and finished cloth shall show not more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.4.

3.4.1 Matching.- The color of the finished cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.4.2 Colorfastness.- The finished cloth shall show fastness to laundering, perspiration, and crocking equal to or better than the standard sample. When no standard sample has been established or designated as applicable to colorfastness, the finished cloth shall show "good" fastness to laundering and perspiration, and shall show a Munsell value for crocking not lower than 8.5. Tests shall be made as specified in 4.4.

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* 3.5 Physical requirements.- The finished cloth shall conform to the requirements specified in table I, when tested as specified in 4.4.

TABLE I.- Physical requirements

Width (exclusive of selvage) minimum	Weight (based on linear yd. of 56 in.)	Yarns per inch minimum		Breaking strength minimum		Weave
		Warp	Filling	Warp	Filling	
Inches	Ounces			Pounds	Pounds	
60	15.0 (min.)	36	36	40	30	2 up 2 down right twill

3.6 Finish.- The cloth shall be fulled (see 4.4) and napped on the face and back equal to the standard sample. The character of nap shall be such as will cause the napped fibers to offer considerable resistance to lifting with a needle.

3.6.1 Acidity (pH).- The pH value of the water extract of the finished cloth shall be no less than 4.0 nor more than 8.0 when tested as specified in 4.4.

* 3.7 Shrink resistant treatment.- The cloth shall be given a shrinkage control treatment for resistance to felting shrinkage by an approved oxidation or resin process (see 6.5). The relaxation shrinkage of the treated cloth shall not exceed 5.0 percent in the warp and 4.0 percent in the filling when tested as specified in 4.4. The felting shrinkage of the treated cloth shall not exceed 4.0 percent in the warp and 3.0 percent in the filling when tested as specified in 4.4.

3.7.1 Alkali solubility.- When an oxidation method of producing shrink resistance is used, the alkali solubility of the treated cloth shall not be increased more than 5.0 percent (absolute) over the untreated cloth when tested as specified in 4.4.

3.8 Length and put-up.- Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 50 yards. Each length shall be put-up on full width rolls as specified in PPP-P-1132.

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3.9 Wool content label.- Cloth manufactured under this specification shall be labeled in accordance with the Wool Products Labeling Act of 1939 (see 2.2 and 4.3.3.3).

* 3.10 Workmanship.- The finished cloth shall conform to the quality established by the specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 First article inspection.- When required, the preproduction sample submitted in accordance with 3.2 shall be visually inspected for appearance, color and finish. The sample shall be tested for physical and chemical characteristics in accordance with table II and 4.4 to the extent applicable.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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4.3.1.1 Examination of wool for grade.- The wool shall be visually examined for grade (see 3.3.1.1) in scoured form (prior to dyeing, picking, or carding) by comparison with the applicable U. S. Standard. The examination for grade shall be performed, separately for each component grade used in the filling blend, on a composite sample of one pound of scoured wool for each 10,000 pounds or fraction thereof in the lot. The composite sample shall be selected from not less than 10 percent of the bales or bags in the lot. The lot shall be unacceptable if any portion of the sample does not conform to the requirements of 3.3.1.1. In the event of dispute resulting from comparison, the grade shall be determined by a width method (wedge).

* 4.3.1.1.1 Reprocessed wool.- Reprocessed wool shall be examined for grade when reduced to the fibrous state prior to blending with other fibers. Method of sampling, sample size and acceptance criteria shall be as specified above (4.3.1.1).

4.3.1.2 Inspection of components and processes.- Components and processes listed in table II shall be tested for the characteristics specified and in accordance with the referenced test methods of CCC-T-191 except where otherwise specified. The lot size shall be expressed in pounds. The sample unit shall be one cone, tube or spool. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. The sample size shall be in accordance with the following table:

Lot size (pounds)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II.- Testing of components

Component	Characteristic	Requirement paragraph	Test method
Nylon	Fiber identification	3.3.1.2	1/
Cotton	Fiber identification	3.3.2.1	1/
Filling yarn	Wool and nylon content	3.3.2.2.	2102
Warp and filling	Yarn ply	3.3.2.1 and 3.3.2.2	Visual 2/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination per sample unit and the result reported as "pass" or "fail".

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4.3.2 In-process.- Inspection shall be made at any point or during any phase of the manufacturing process to assure that only approved wools have been used (see 3.3.1.1). The Government reserves the right to exclude from consideration for acceptance any cloth for which the in-process inspection has indicated nonconformance.

4.3.3 Examination of end item.- Examination of the end item shall be in accordance with 4.3.3.1 through 4.3.3.4.

4.3.3.1 Yard-by-yard examination.- Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (rolls) 1/
3200 or less	8
3201 up to and including 10,000	13
10,001 and over	20

1/ No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls in which case all shipping containers shall be present in the sample.

All defects, as defined in Section II of Fed. Std. No. 4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.3.3.1.1. The same yardage shall be given a through-lighting inspection from the back of the cloth towards a light source for holes and thin areas not previously scored on the face side of the cloth. Slight open places resulting from removal of burrs, shives, sisal or broken picks not previously scored on the face side o shall not be scored in through-lighting inspection. Defects on the selvage shall not be scored unless the body of the cloth is affected. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 10.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 15.0 points. If no individual roll exceeds 15.0 points per 100 square yards, the lot shall be acceptable with respect to this characteristic. If one roll exceeds 15.0 points per 100 square yards, a second sample shall be examined only for individual roll quality

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examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 15.0 points per 100 square yards. For individual roll quality examination, local type defects only shall be scored. Overall type defects shall be scored only in determining compliance with the lot average quality requirements. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.3.3.1.1 Demerit points.- Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	One point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension . . .	Two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension . . .	Three points
For defects exceeding 9 inches in any dimension	Four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Nap offers only slight resistance to lifting with a needle.
Degree and character of finish on face and back not equal to standard sample.
Mottled, cloudy or streaky.
Width less than specified.
Baggy, ridgy or wavy cloth.
Off shade; shaded side to side, side to center, or end to end.

4.3.3.2 Examination for length.-

4.3.3.2.1 Individual rolls.- During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective with respect to length.

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4.3.3.2.2 Total yardage in sample.- The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the roll tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.3.3.3 Examination for compliance with Wool Products Labeling Act.- During the yard-by-yard examination, each roll in the sample shall be examined for conformance to the Wool Products Labeling Act. Each roll not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.3.3.4 Examination of preparation for delivery requirements.- Examination shall be made in accordance with the provisions of PPP-P-1132 to determine that packaging, packing, and marking requirements of Section 5 of this specification are complied with.

* 4.4 Tests.- The methods of testing specified in CCC-T-191 wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the result(s) of the determination(s) made on a sample unit for test purposes as specified in the applicable test methods. The sample unit shall be 1/4 yard untreated and 3-1/2 continuous yards, full width, of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The lot size shall be expressed in units of one yard. All test reports shall contain the individual values utilized in expressing the final result. The sample size (number of sample units) shall be in accordance with the following:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III.- Test methods

Characteristic	Requirement reference	Test method
Presence of labile sulfur	3.4	2020
Colorfastness to:		
Laundering	3.4.2	5614
Perspiration	3.4.2	5680
Crocking	3.4.2	5651

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TABLE III.- Test methods (Cont'd)

Characteristic	Requirement reference	Test method
Weight	3.5	5041
Yarns per inch:		
Warp	3.5	5050
Filling	3.5	5050
Breaking strength:		
Warp	3.5	5100
Filling	3.5	5100
Weave	3.5	Visual 1/
Finish		
Fulled on face and back	3.6	2/
Acidity pH	3.6.1	2811
Shrinkage in relaxation:		
Warp	3.7	5558
Filling	3.7	5558
Shrinkage in felting:		
Warp	3.7	5554 3/
Filling	3.7	5554 3/
Alkali solubility	3.7.1	2800 4/

1/ One determination per sample unit and the result reported as "pass" or "fail".

2/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

3/ The felting shrinkage (5554) shall be determined on the same specimens used for relaxation shrinkage (5558) and determined by the following formula:

$$\text{Felting shrinkage percent} = \frac{B - F}{B} \times 100$$

Where B = Average measurement after relaxation.
Where F = Average measurement after felting.

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4/ Two treated and two untreated specimens shall be tested per sample unit.

* 5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C.- The cloth, put-up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1132.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C.- The cloth shall be packed in accordance with the applicable requirements of PPP-P-1132.

5.3 Marking.- In addition to any special marking required in the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1132.

6. NOTES

6.1 Intended use.- The cotton warp and wool-nylon filling cloth is intended for the lining for fur ruffed winter hood.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) First article - When preproduction sample is not required (see 3.2).
(c) Minimum length, if other than specified (see 3.8).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample.- For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 The standard sample, Olive Green 104, has been dyed with the following formulation of dyes. This is for information only and does not constitute a mandatory requirement.

Direct Black #28 C.I. 30235
Direct Red #83 C.I. 29225
Direct Orange #42
Neutral Premetallized Olive B GL
Neutral Premetallized Yellow 2 RL

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* 6.5 Only those shrink-resistant chemical treatments already approved by the appropriate medical service and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement (see 3.7).

6.6 Marginal notations.- The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever, for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Army - GL

Review activity:

Army - MD

Preparing activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION Cloth, Fleece, Cotton Warp and Wool-Nylon Filling: Lining 15-Ounce, Shrink Resistant MIL-C-2069E(GL)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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